

LECTURE 3 — COAL COUNTRY AND THE BATTLE OF BLAIR MOUNTAIN

PHY199-RAP · Energy in the 21st Century

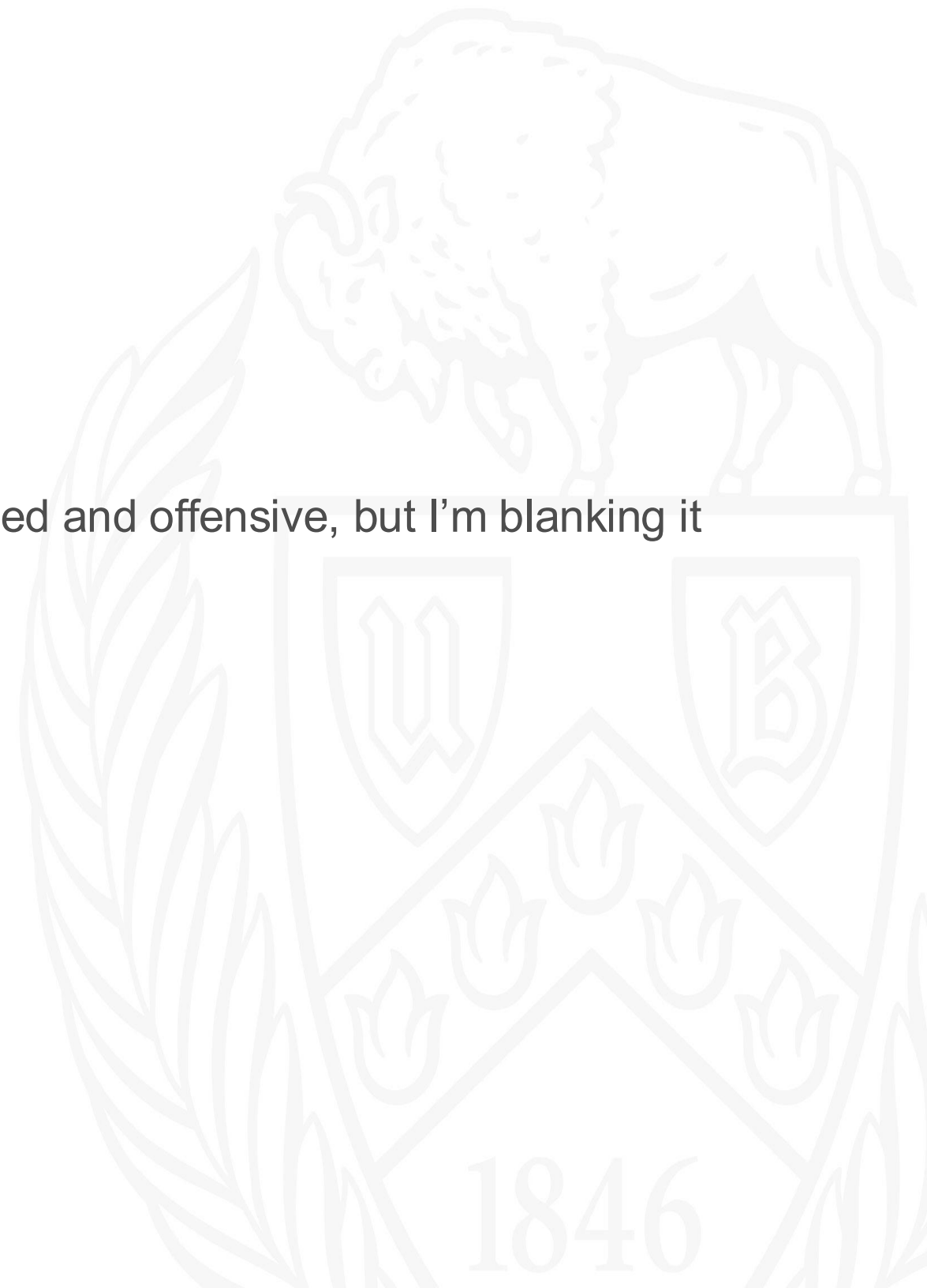
Dr. Salvatore Rappoccio

 **University at Buffalo** The State University of New York

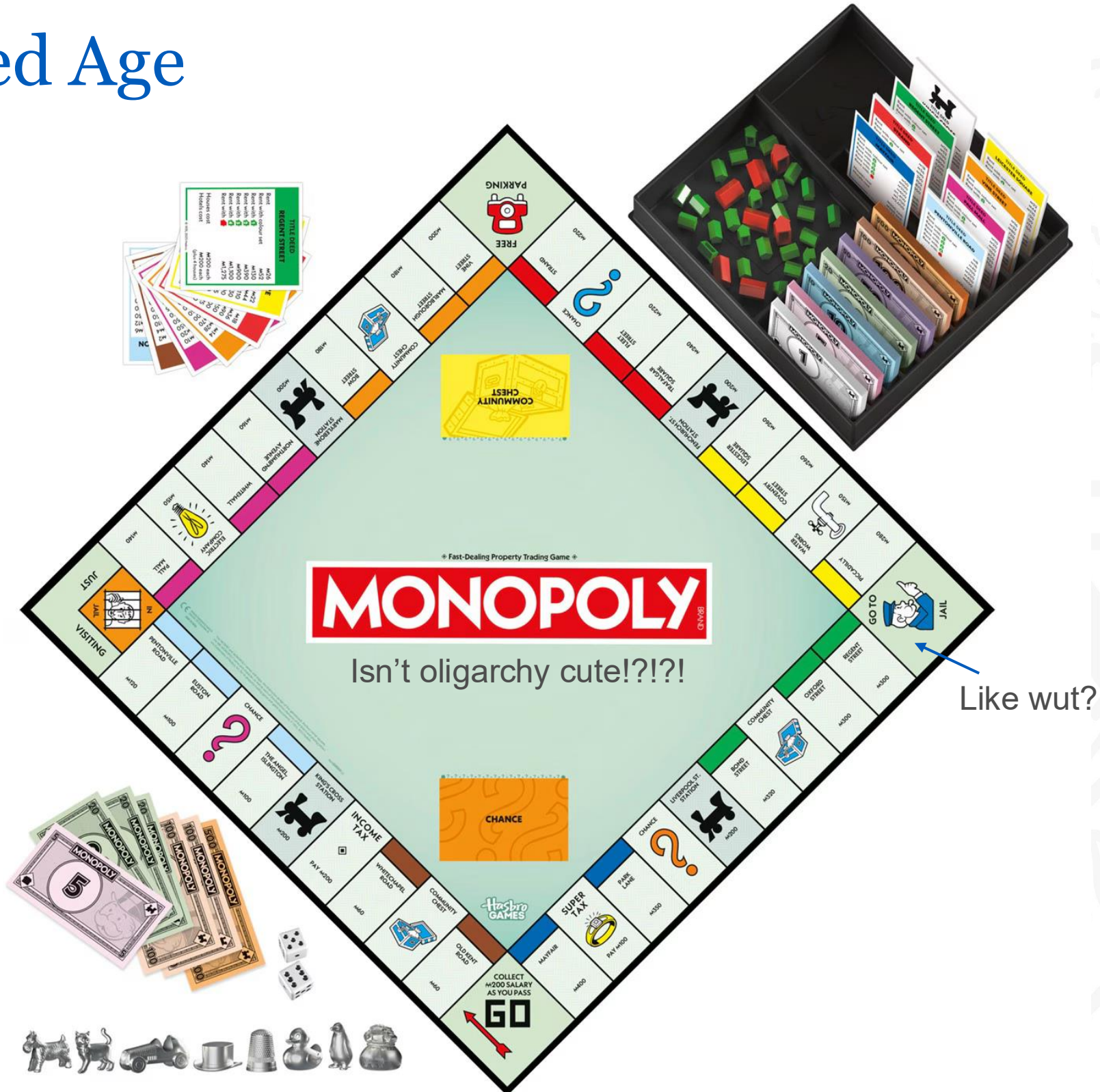
1846

Content Warnings

- We will be discussing
 - The Silk Road
 - European Colonialism
 - Genocide of Native American people
 - The enslavement of African people (language is dated and offensive, but I'm blanking it out)
 - War and deaths
 - Child endangerment and labor



The Gilded Age



AI Report

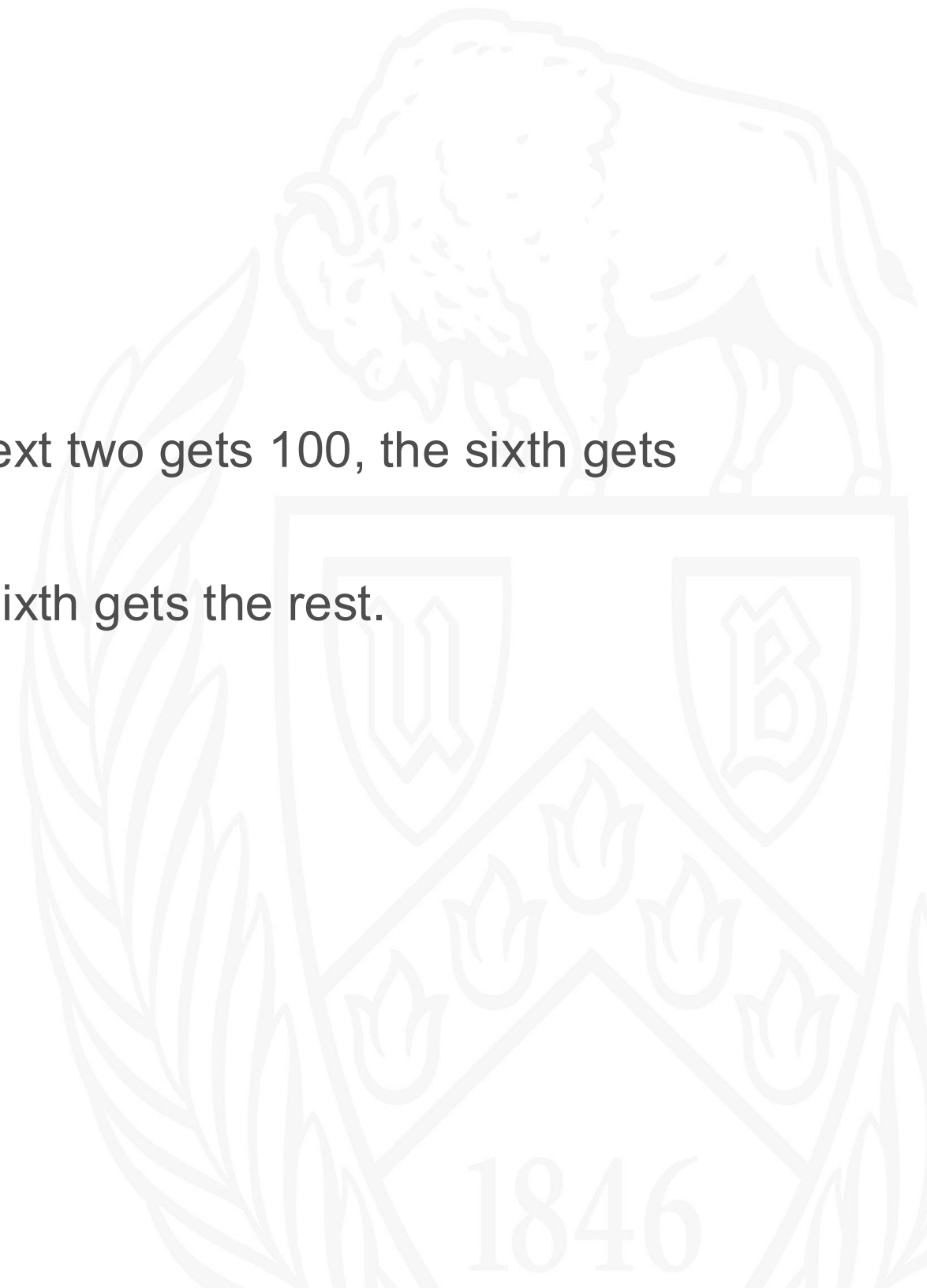
- I used claude code to generate a Monopoly game, took about 2 hours and 20M tokens
 - My prompt:
 - I want to build a trivial Monopoly game, but with a modification. Read the twiki here to refresh yourself on the rules:
 - <https://www.wikihow.com/Play-Monopoly>
 - I want it to be python-only.
 - Make it a pygame engine.
 - The rules should be as-written, but instead of each player getting \$1500, the amounts are set ahead of time by a game manager.
 - Give me a markdown plan and we will discuss.
 - Then for each step: “Read PLAN.md and implement Phase X. Make unit tests to ensure they keep working throughout the game. At the end, when the tests are green, tell me and we will mark it done on the PLAN.md file, and I will refresh.”
 - There was occasional fine-tuning that I won’t list
- Let’s play!

Plan.md

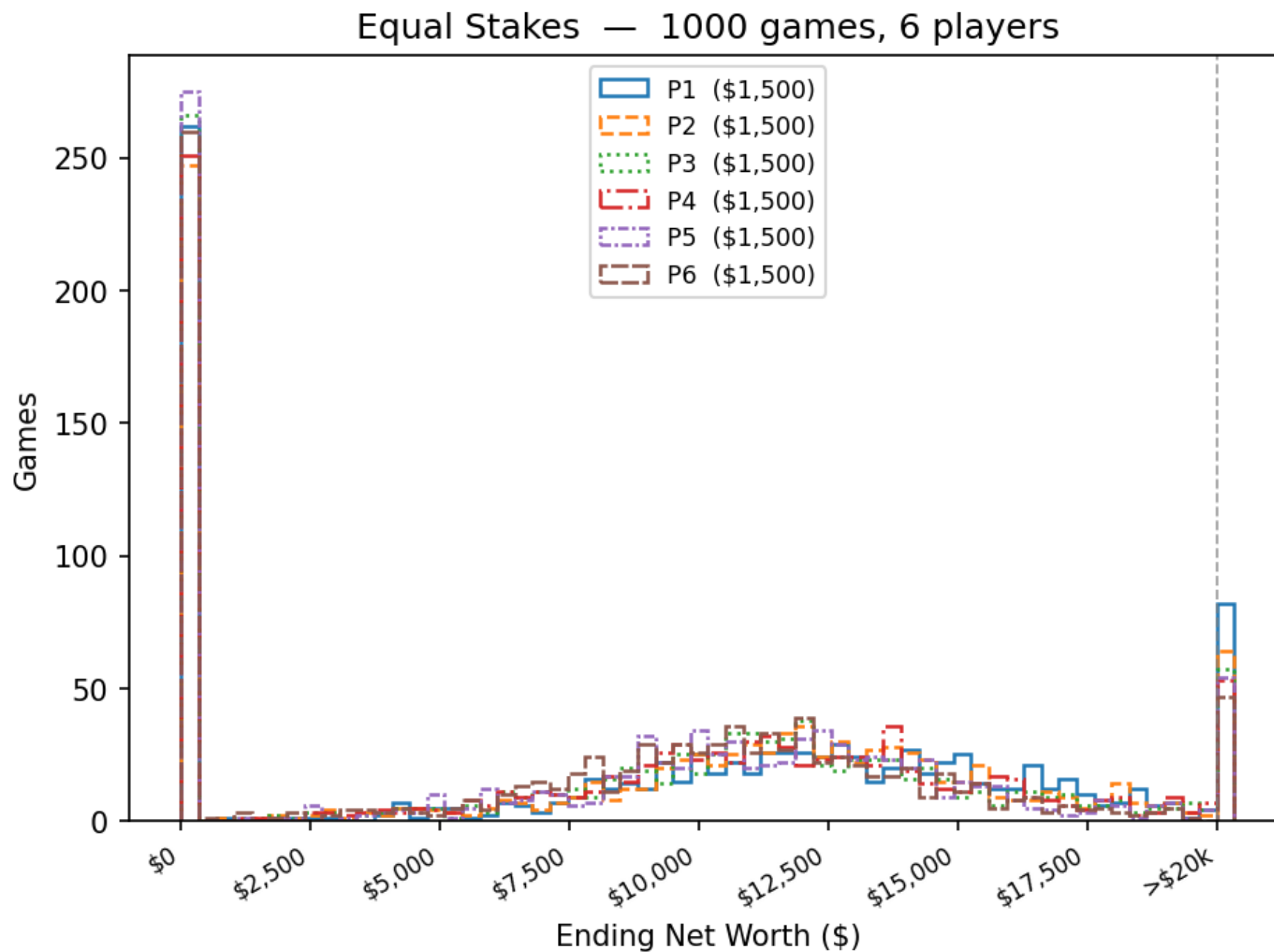


Data Analysis

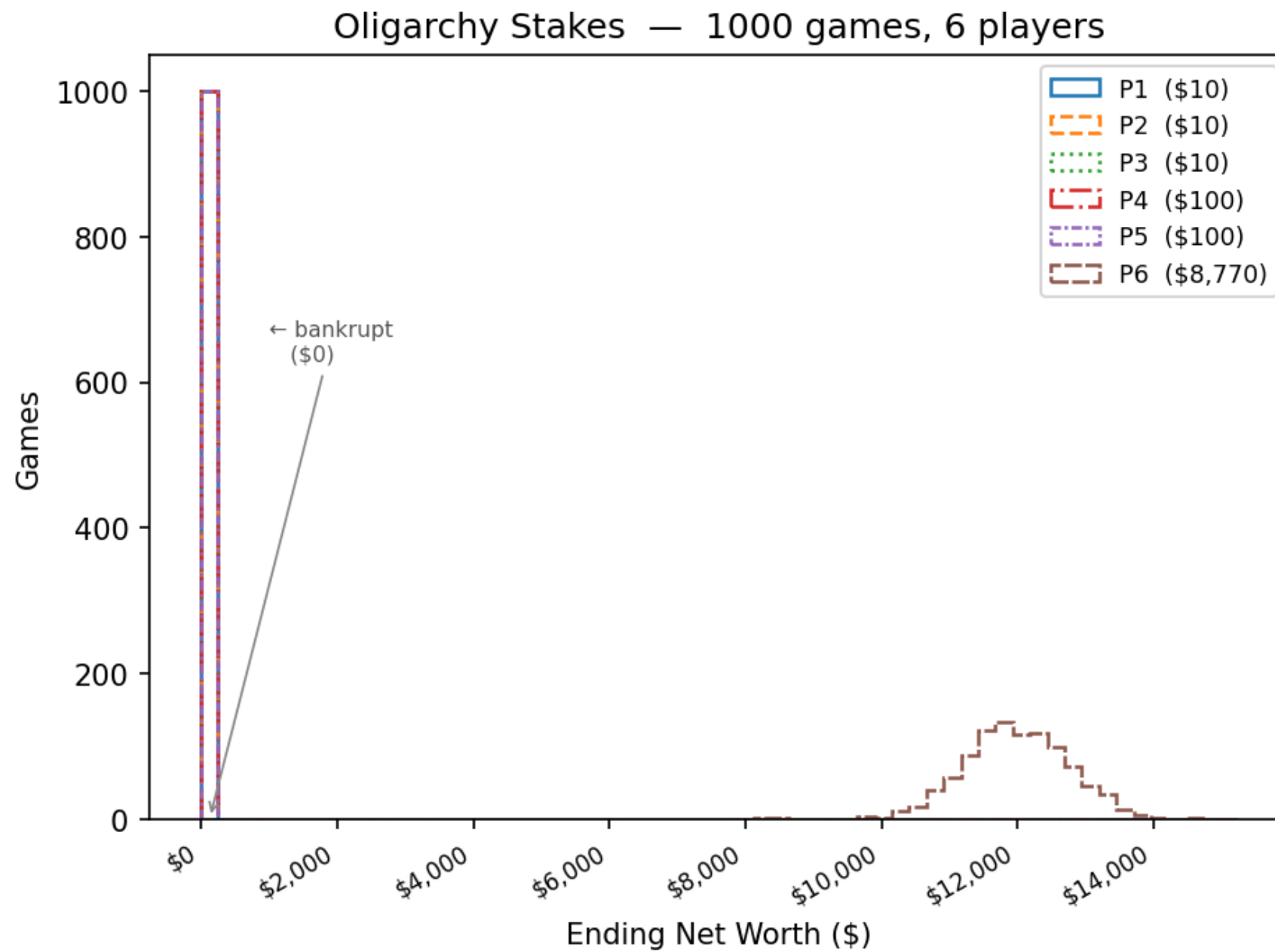
- Looked at three configurations:
 - Equal stakes: All six at 1500
 - Oligarchy stakes: lowest three each get 10, next two gets 100, the sixth gets the rest.
 - Company town stakes: lowest five get 1, the sixth gets the rest.



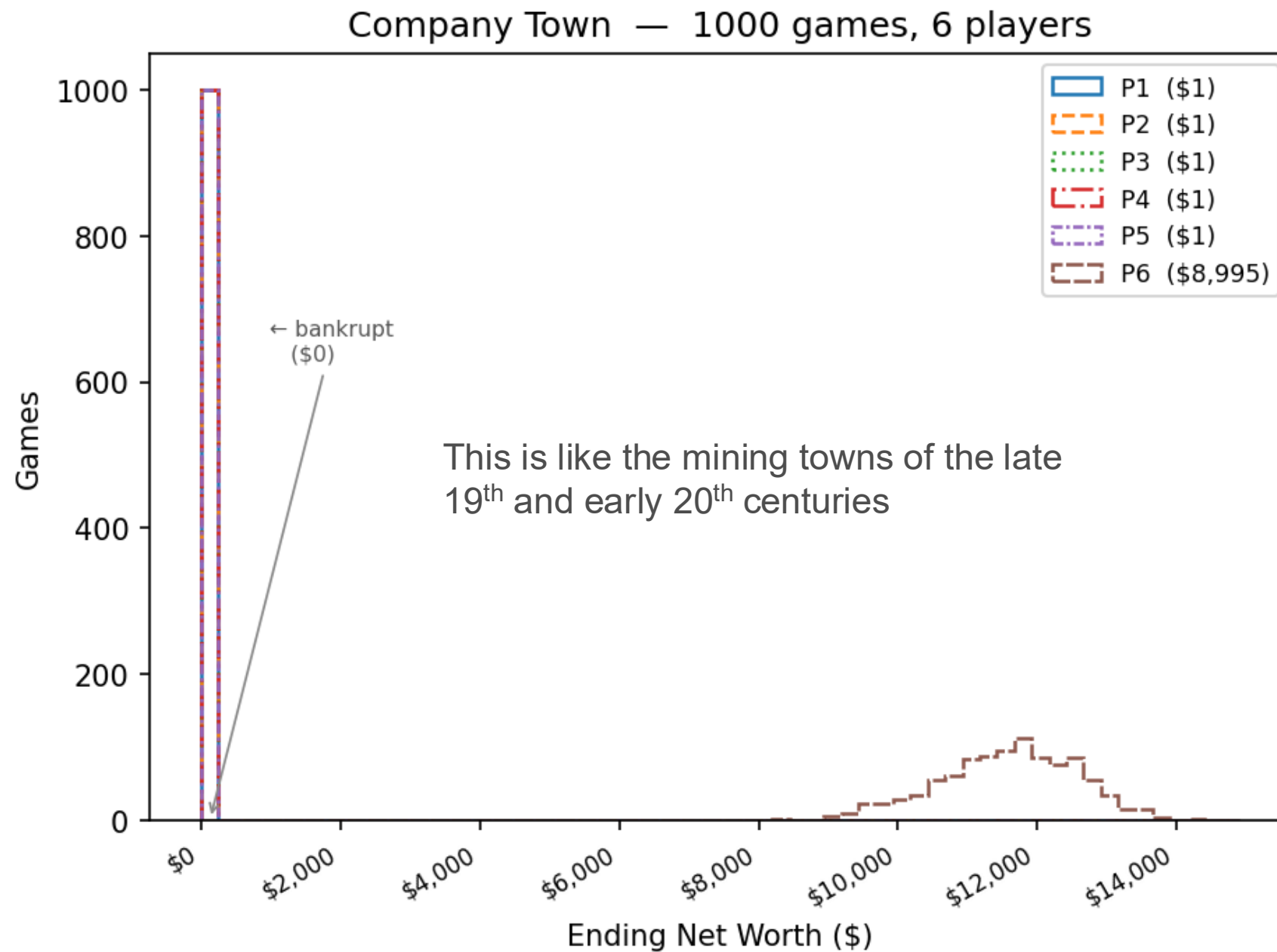
Data Analysis



Data Analysis

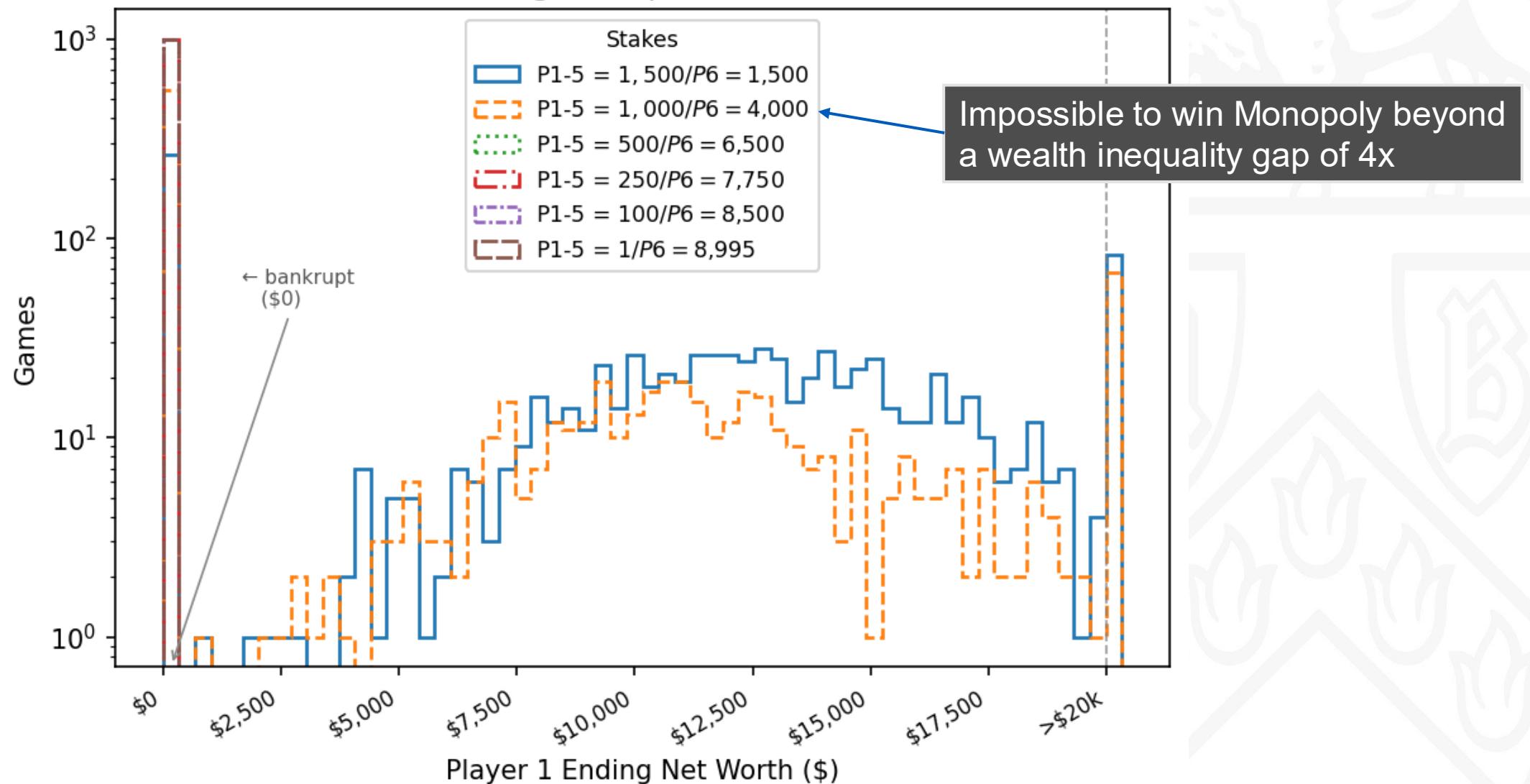


Data Analysis



Data Analysis

Player 1 outcomes as stake advantage shifts to P6
1000 games per scenario



THE GLOBAL CONTEXT



Global Context

- [Europeans liked Asian stuff](#) in the Medieval period (Silk, Spices, Ivory)
- [Marco Polo](#) (1254-1324) was famous for traveling the (existing) [Silk Road](#)
- The Roman Empire (Eastern version) fell in 1453 during the [Fall of Constantinople](#) to [Mehmet II](#) of the much more advanced [Ottoman Empire](#) (who used cannons to destroy the walls of the city)
- Europe had to (gasp!) now pay taxes for their favorite stuff, so they traveled westward trying to get to Asia

https://en.wikipedia.org/wiki/Marco_Polo



Saffron is
yummy

https://en.wikipedia.org/wiki/Silk_Road



https://en.wikipedia.org/wiki/Mehmed_II



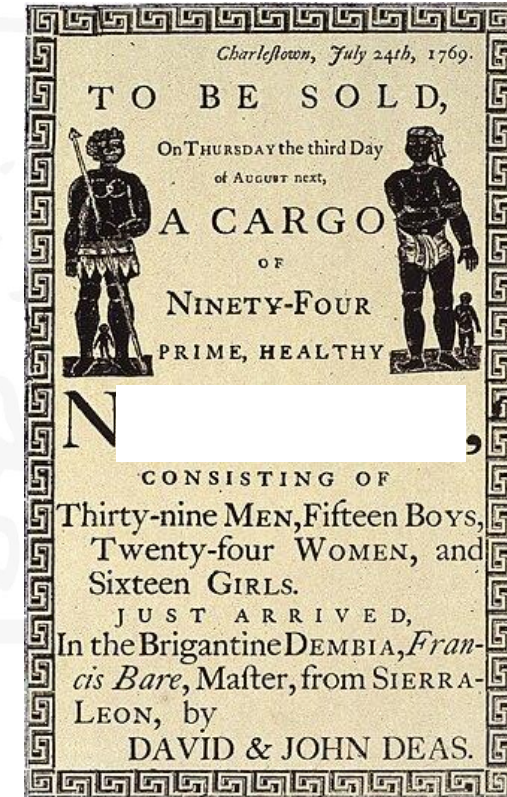
Lol get
wrekt,
pay me

Global Context

- 1492, Columbus convinces Queen Isabella I and King Ferdinand II to finance his trip to Asia
 - Everyone knew the earth was a sphere (since around 500 BCE)
 - Finds a continent in his way, genocides the people there
- Sugar plantations become hugely profitable, but the Native Americans were dying from conquest and diseases they had no immunity to
 - Thus Europeans kidnapped people from Africa to make the global economy run
- Eli Whitney's Cotton gin (1793) put the enslavement of people into overdrive
- The US then went to war with itself because the North decided that maybe we shouldn't enslave people, but the plantation owners in the South decided they had enough



https://en.wikipedia.org/wiki/Christopher_Columbus



https://en.wikipedia.org/wiki/Atlantic_slave_trade



https://en.wikipedia.org/wiki/American_Civil_War

“Captains of Industry”

Who	What they controlled	The number
Cornelius Vanderbilt (1794-1877)	Steamboats, then railroads; consolidated the New York Central, 1870	Estate at death, 1877: \$105 million (~\$3.2 B in 2025 dollars)
John D. Rockefeller (1839-1937)	Standard Oil, founded 1870	~ 90% of US oil refining by 1900; worth \$900 M in 1913 = 2.3% of US GDP
J. P. Morgan (1837-1913)	Banking; railroads; General Electric; U.S. Steel	U.S. Steel, 1901: the world's first billion-dollar company (\$1.4 B)

The House of Morgan

Miles Morgan (1616-1699)

Nathaniel Morgan (1671–1752)

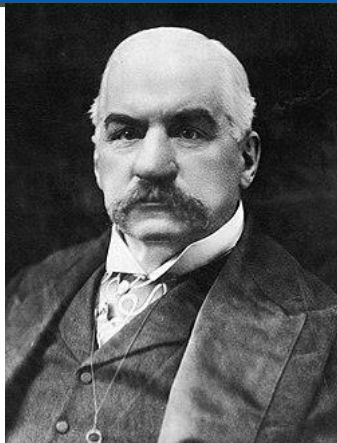
Joseph Morgan (1702-1786)

Joseph Morgan Jr. (1736–1813)

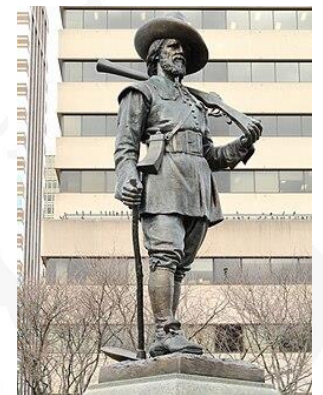
Joseph Morgan III (1780–1847)

Junius Spencer Morgan I (1813-1890)

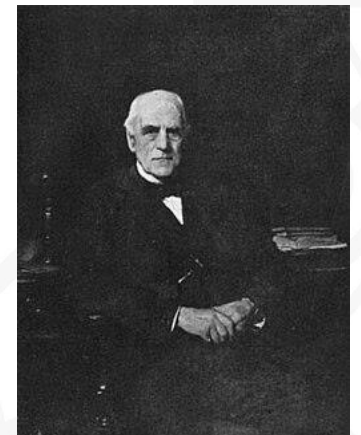
John Pierpont Morgan Sr (1837-1913)



- Colonizer from Wales, moved to [Massachusetts Bay Colony](#) in Roxbury
- Moved to Springfield, defended it against the [Nonotuck nation](#) who attacked in retaliation to the colonists forcibly disarming them, and subsequently trying to reclaim their ancestral homelands in the [Siege of Springfield](#), part of [King Philip's War](#) (aka [Metacomet's](#) war)



- Fought in Revolutionary War
- Got involved in banking
- Further involvement in banking, including [financing the slave trade and plantations](#)



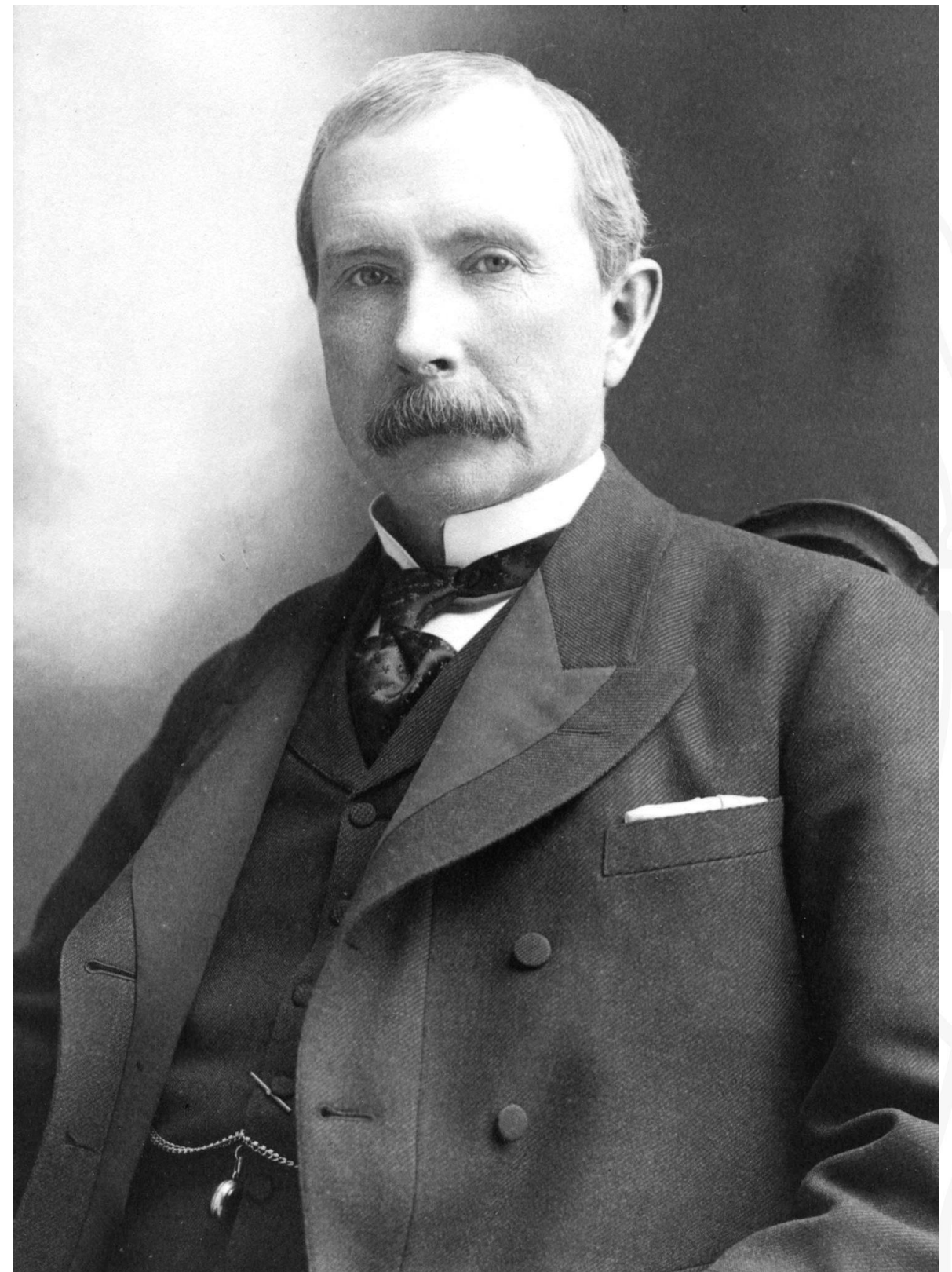
- Revolutionized financing
- Already met him in the War of the Currents
- He was a 6th generation nepo baby and inherited the equivalent of ~\$400M, then grew to ~\$2B

1846

1865-1900: Railroad expansion, genocide, and industrialization

After the Civil War, the US began a huge expansion west and rapid industrialization

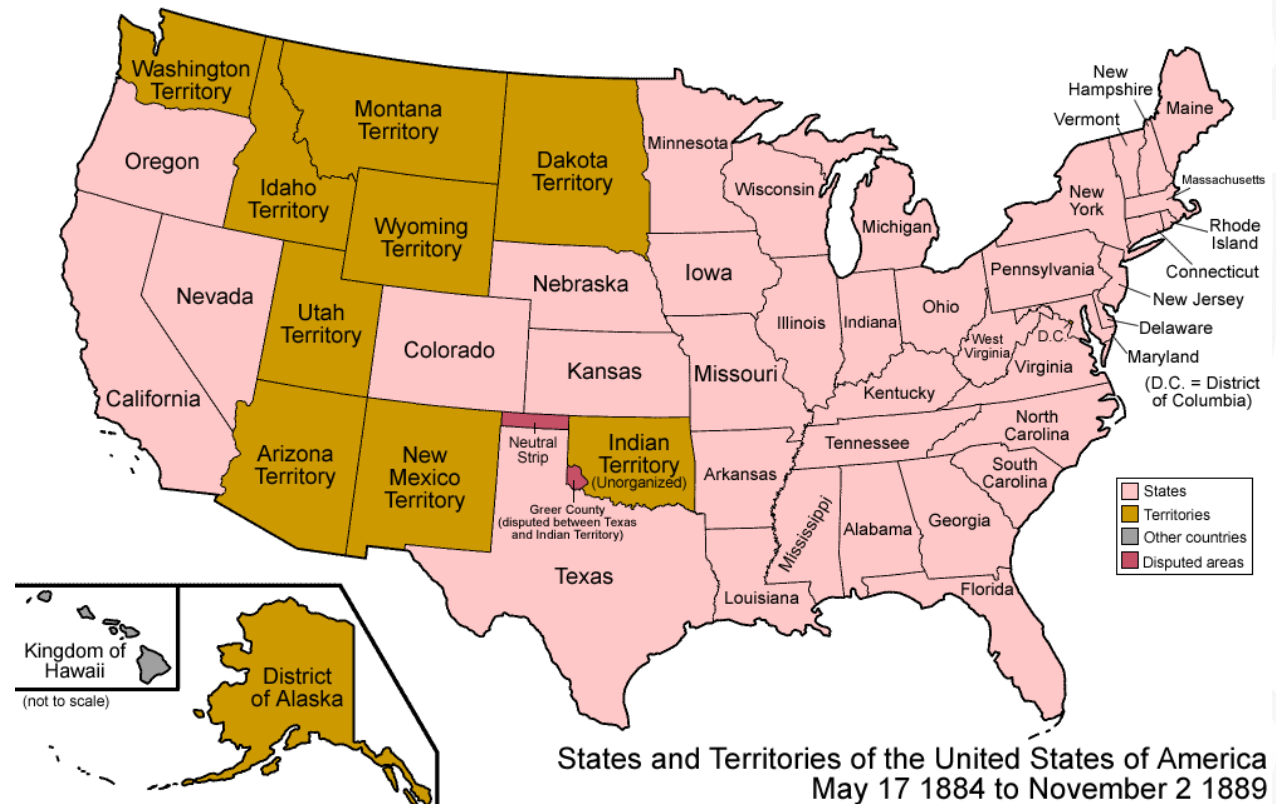
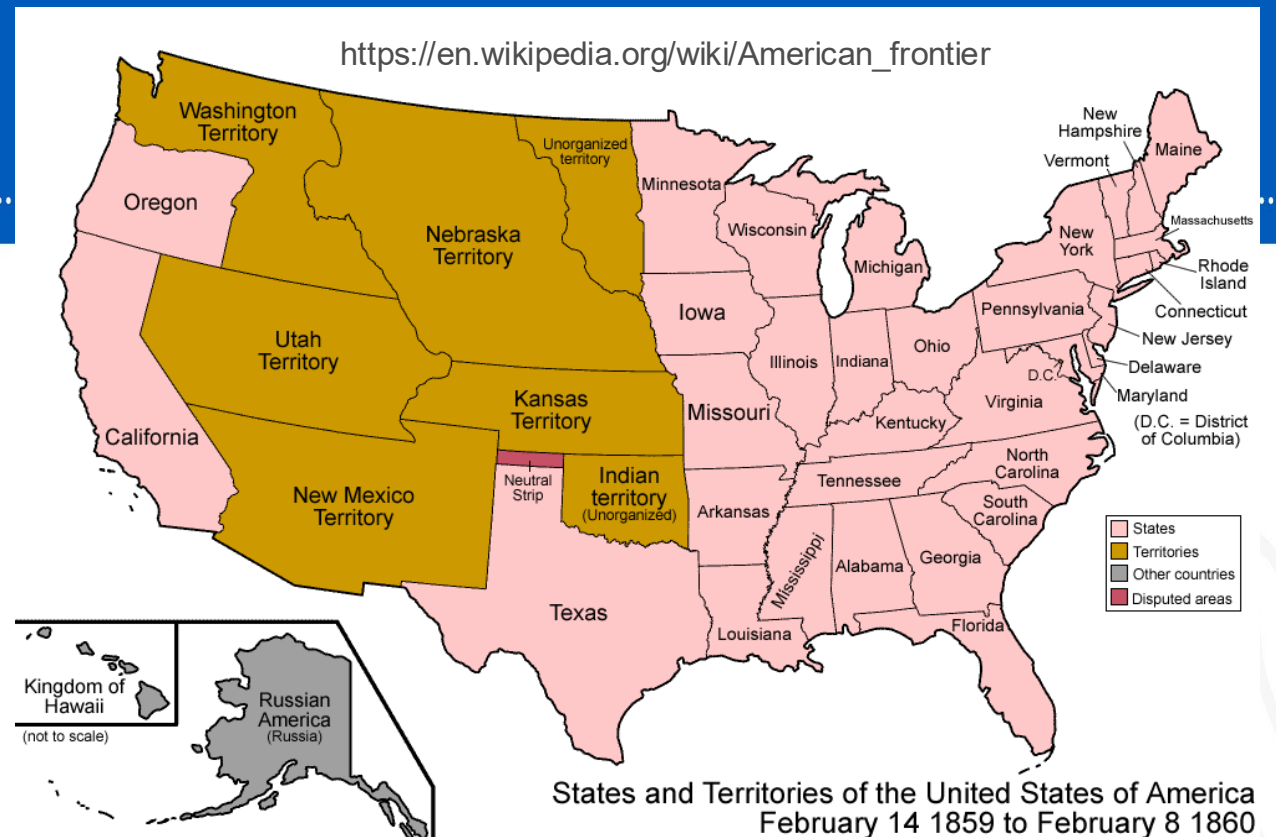
- **Railroads** began to be built, and needed steel, coal, and inexpensive labor
 - [Baltimore and Ohio Railroad \(B&O\)](#), [Reading Railroad](#), [Pennsylvania Railroad](#), [Mohawk & Hudson Railroad](#), [Saratoga & Schenectady Railroad](#), [Lehigh Coal & Navigation Company](#), [Tuscumbia, Courtland & Decatur Railroad](#)
- **Steel** becomes cheap (Bessemer) — so more rail, more bridges, more mills
- **Oil** becomes a commodity: Rockefeller founds **Standard Oil of Ohio**, 1870
- **Electricity** becomes an industry: Edison's Pearl Street station, 1882 — *where Part 1 started*



John D. Rockefeller (1839-1937). Public domain.

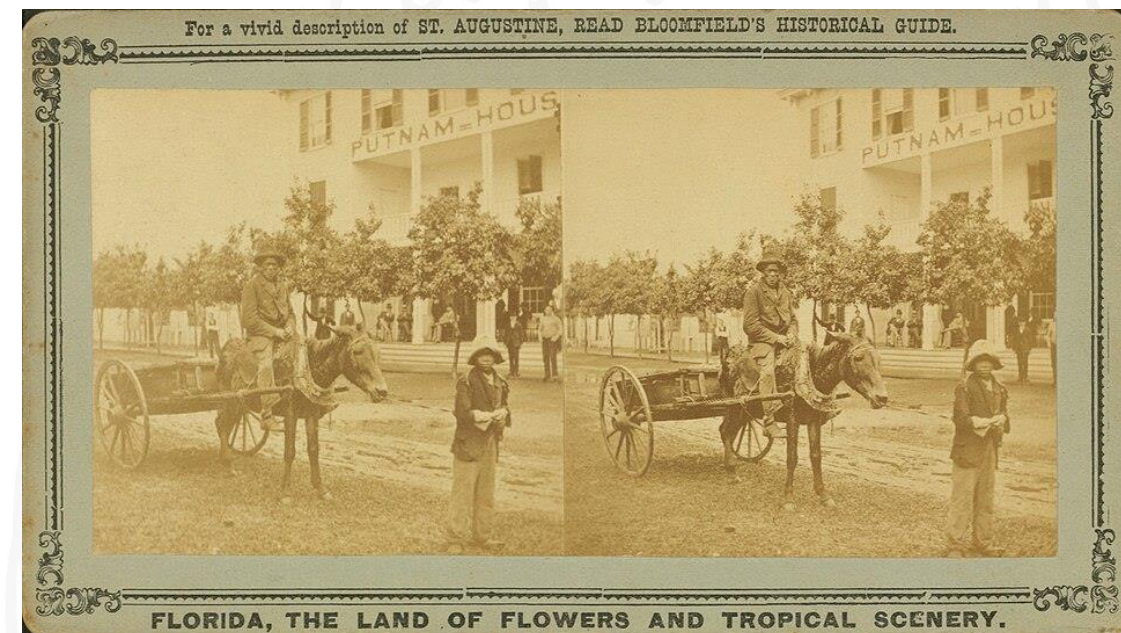
Genocide in Hyperdrive

- The 19th Century resulted in massive expansion westward into Native American territories and Mexican territories and subsequently resulted in a massive genocide of [Native Americans](#).
 - [Trail of Tears](#), [Long Walk](#), [Yavapai Exodus](#), [California Genocide](#), etc, creation of the [reservation system](#) that we continue to exploit today
- The railroad allowed the US to move very quickly and transport new weaponry developed in the Civil War to exterminate most of the Native American population in the US



Reconstruction, Jim Crow, and the Great Migration

- After [Sherman's March to the Sea](#), Sherman promised enslaved Black people [40 acres and a mule](#) after the war
- Similarly, the US government had passed the Homestead Act in 1861, promising the stolen land from Native American and formerly Mexican land via the [Homestead Act](#)
- After [Lincoln](#) was murdered, [Andrew Johnson](#) reneged on the promise, refused to prosecute the leaders or members of the Confederacy, and [the land given to the formerly enslaved Black people was given back to the slave-owning plantation owners](#), reneging on the promise
- This began the era of [sharecropping](#)

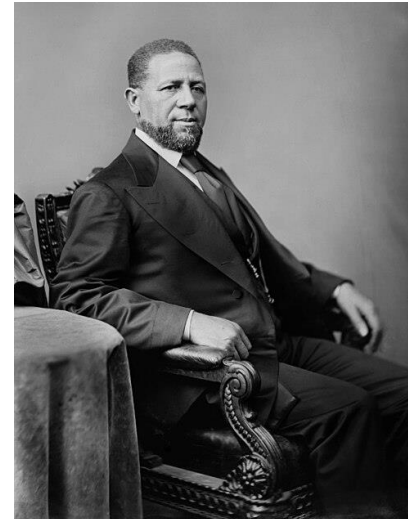


https://en.wikipedia.org/wiki/Forty_acres_and_a_mule

Reconstruction, Jim Crow, and the Great Migration

- After the war, Black Americans started winning elections for US Congress
 - Hiram Revels (Senate)
 - Joseph Rainey (House)
- The white Americans in power then implemented brutal suppression tactics of Black people
 - Jim Crow laws
 - Atlanta 1906 race massacre
 - Devastation of Tulsa, Oklahoma's "Black Wall Street"

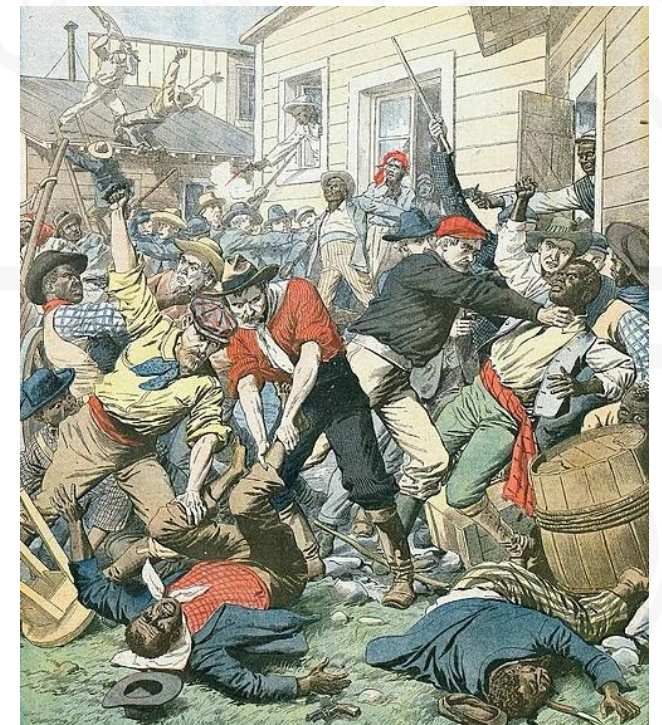
Hiram Revels



Joseph Rainey



Cover of Le Petit Journal 7-Oct-1906 depicting the massacre of Atlanta

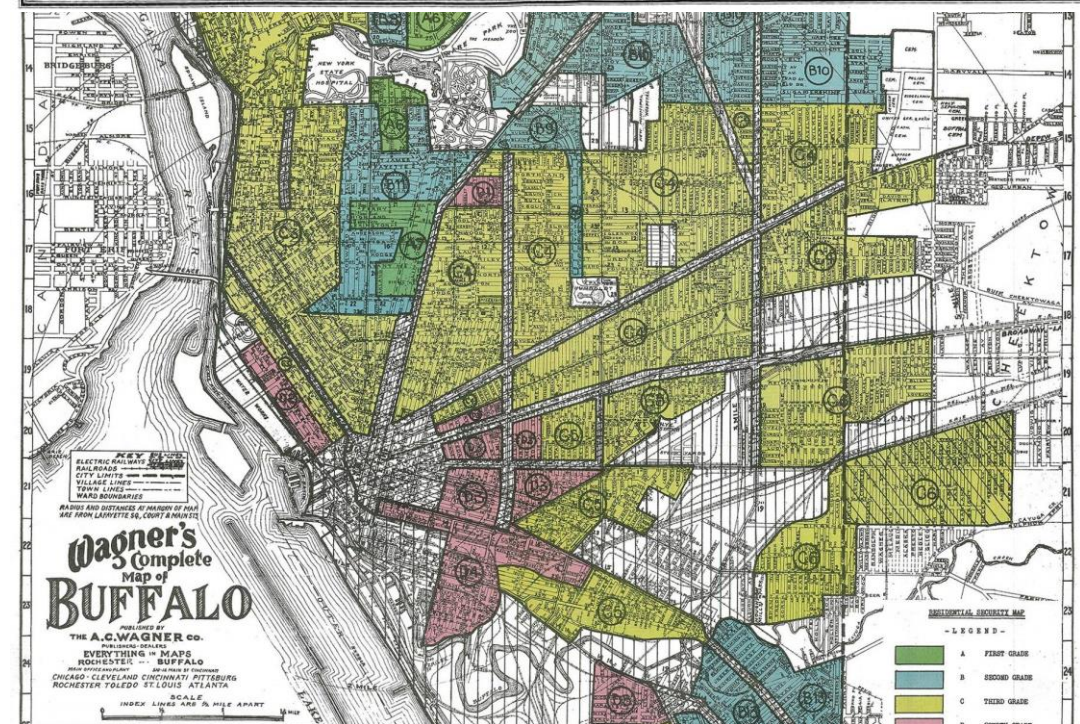
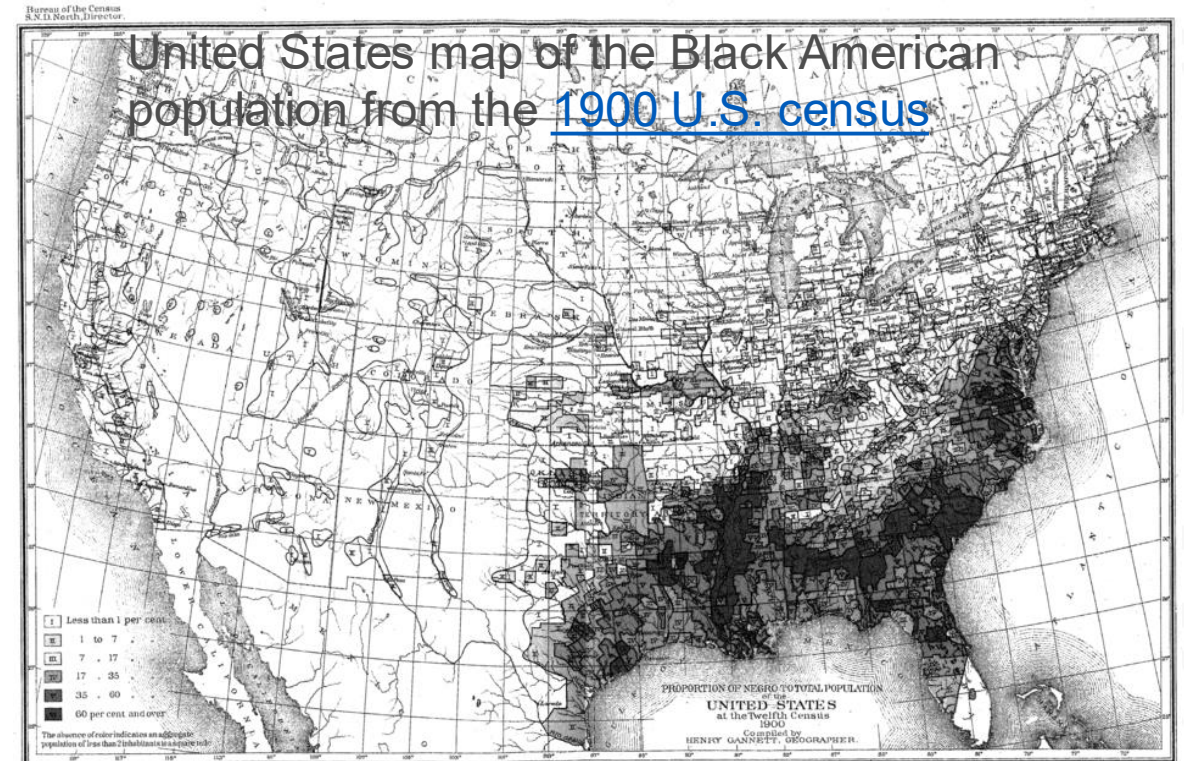


https://en.wikipedia.org/wiki/Jim_Crow_laws

Reconstruction, Jim Crow, and the Great Migration

- This and other factors led to the [Great Migration](#) of Black Americans from the South to other areas
- Also led to extensive [redlining](#), of which [Buffalo is one of the worst in the Northeast](#)

These people had no net wealth, and the US took away what was promised to them

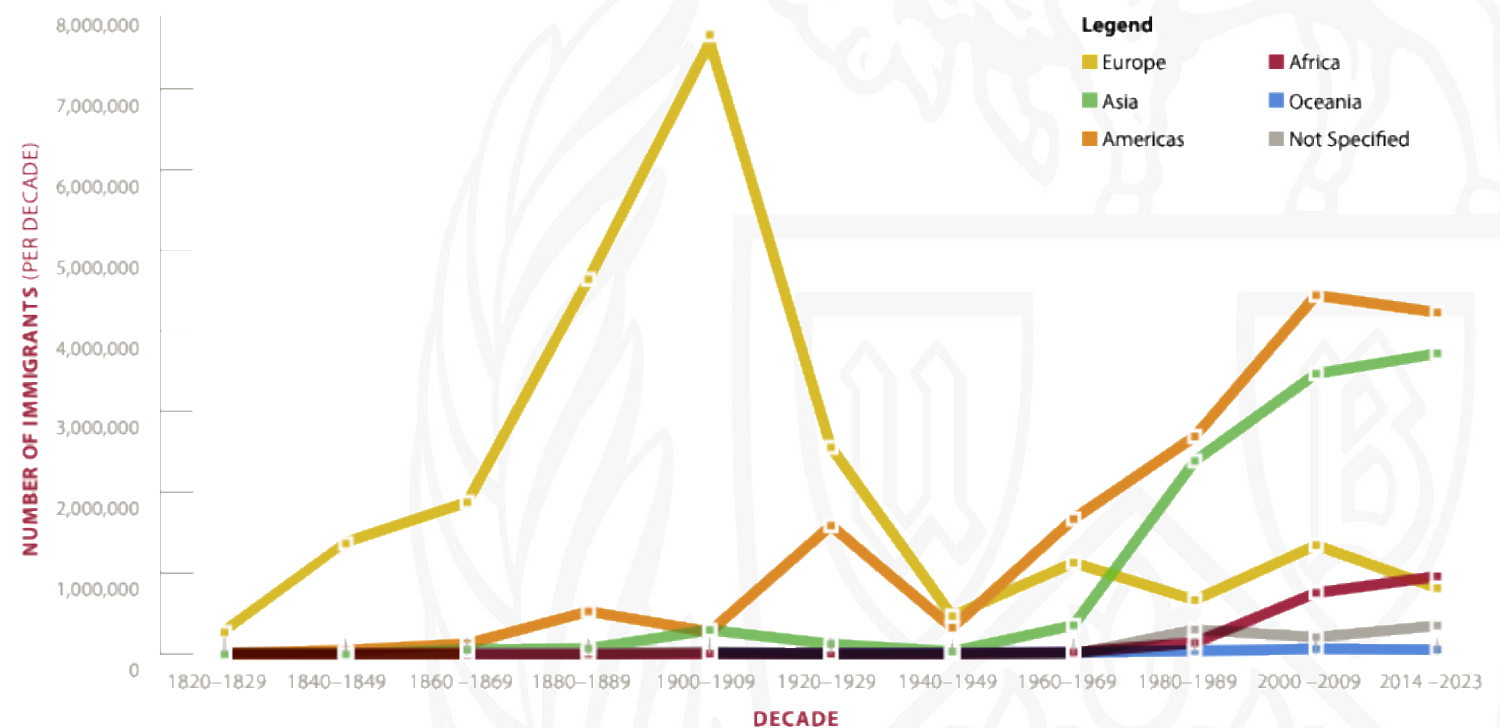


Massive Immigration

- [Asian](#), [American](#), and [European](#) immigration to the US surged in the late 19th and early 20th centuries
 - For American immigration, often the US immigrated to THEM, not the other way around, after the [Mexican American](#) war
 - [Irish Potato Famine](#)
 - Unifications of [Germany](#) and [Italy](#)
 - [Jewish people escaping Pogroms in Eastern Europe](#)



Immigration to the U.S. by Geographic Region (1820–2023)



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Source: U.S. Department of Homeland Security

330+
MILLION
IN THE
USA

Almost none of these people had any net wealth at all

Conditions for Exploitation

- Extensive violence against and brutal suppression of Native, Black, Asian, Latinx, and European immigrant Americans
- Desperate workforce, living on subsistence
- New technology that led to enormous wealth acquisition
- Extensive new capital and banking
- Newly discovered resources that needed to be extracted
- No regulation or oversight from government

Let's guess
what happened

Sources

- **Wikipedia entry points** (finding aids — trace to the primary sources they cite): Battle of Blair Mountain · Coal Wars · Company town · Coal Region · Susquehannock · Ludlow Massacre · Coal strike of 1902 · Muckraker · Starbase, Texas · Marco Polo (Silk Road travel) · Silk Road (medieval trade route) · Mehmed II (Ottoman conquest) · Christopher Columbus (1492 voyage) · Atlantic slave trade (African enslavement) · American Civil War (North vs. South) · American frontier (westward expansion) · Forty acres and a mule (Reconstruction promise) · Jim Crow laws (racial suppression)
- **Institutional / scholarly:** West Virginia Mine Wars Museum · *e-WV: The West Virginia Encyclopedia* (scrip) · Smithsonian Magazine, "What Made the Battle of Blair Mountain the Largest Labor Uprising in American History" · Price V. Fishback, *Journal of Economic History* · Scott Reynolds Nelson, *Steel Drivin' Man* (2006) · Lon Savage, *Thunder in the Mountains* (1990)
- **Contemporary reporting:** Texas Tribune (Starbase incorporation; HB 5246) · Scientific American (Starbase opposition) · New Republic ("Freedom Cities")
- **Watch / read:** *Matewan* (dir. John Sayles, 1987) · Denise Giardina, *Storming Heaven* (1987)